
MLPC Report - Task 3: Data Exploration

Team Temporal Frameworks

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1 Verify Annotations & Quality Analysis

1.1 Summary of Disagreements

Meaningful disagreements mostly arise from misunderstandings or from the fact that some of the agreements themselves are not clearly defined. For example in Figure 1, the “one-second rule” is sometimes interpreted too strictly or too loosely.

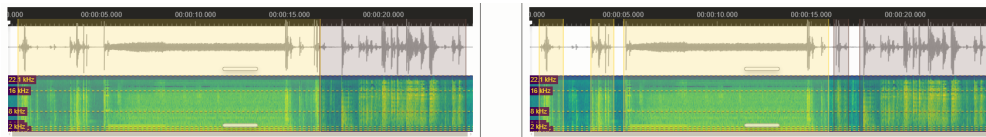


Figure 1: The right annotator used more breaks in between annotated areas than the left.

The breaks between annotated segments are sometimes shorter and sometimes longer than one second. Whether these breaks should be included depends on how the situation is interpreted. One possible perspective is that only one dish being audible at a time defines the segment, while another is that the broader context of the dishes also plays a role.

1.2 Case Study

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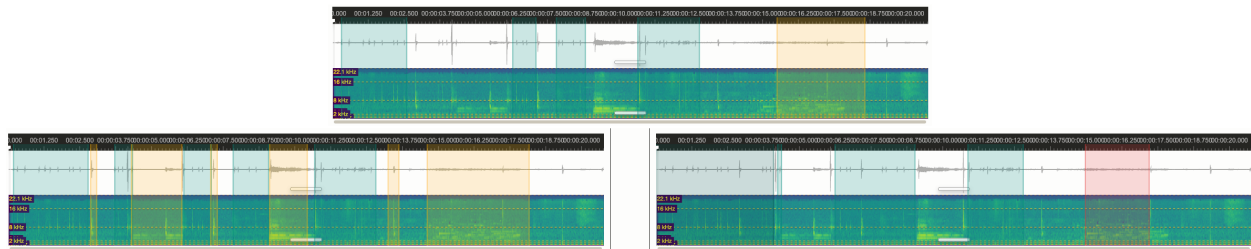


Figure 2: Side-by-side comparison of Annotator 1 (top), Annotator 2 (left) and Annotator 3 (right), showing heavy annotation disagreements.

This case highlights strong disagreements caused by flawed metadata provided by the creator of the recording. The metadata listed `notification_sounds` as a non-target class, contradicting the official guidelines which state these must be labeled as `phone_ringing`.

As shown in Figure 2, Annotator 1 and 3 followed the incorrect metadata, missing all phone notification sounds. Annotator 3 further violated the guidelines by labeling the final incoming phone call as `bell_ringing` (strictly

reserved for doorbells). In contrast, Annotator 2 caught the error and labeled the notifications as `phone_ringing`, but also included the outgoing "message sent" sounds, suggesting ambiguity in the class boundaries.

Ultimately, relying on the provided descriptions over official definitions introduces a systematic bias, leading to missing or invalid labels. It also emphasises the need to clarify if active usage sounds, like sending a text, belong in the `phone_ringing` class.

2 Annotations

2.1 Annotator Agreement

Looking at the Jaccard similarity across all files and classes, annotators tend to agree on roughly 70% of the labels. We compute this by binarising all annotated regions and dividing the intersection of all annotations by the union — basically measuring how much everyone overlaps in their labeling.

Some classes have much higher agreement. For example, `vacuum_cleaner` and `keyboard_typing` reach about 82% and 85%. This makes sense because these sounds are pretty distinct: vacuum cleaners are loud and keyboards are usually used in quiet settings, so both are easy to hear clearly.

Other classes, like `wardrobe_drawer_open_close`, `door_open_close`, and `footsteps`, have lower agreement, around 50%. The low consensus might be because these events occur frequently or are more subtle, making them harder to annotate consistently — especially footsteps, which appear all over the dataset.

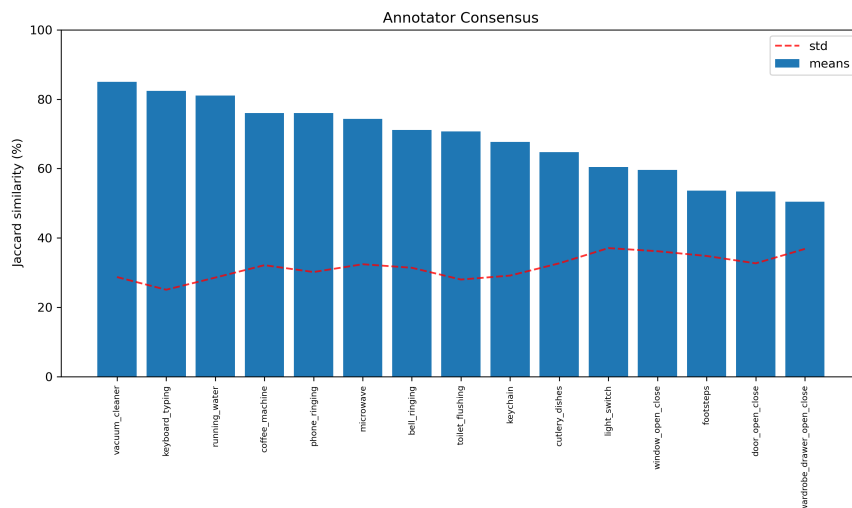


Figure 3: Annotator agreement across classes. Bars show the mean Jaccard similarity, and the red line represents standard deviation.

Overall, the relatively low standard deviation values indicate that annotator agreement is fairly consistent, even if not perfect.

As expected, we observe an inverse relationship between mean Jaccard similarity and its standard deviation: as similarity decreases, the standard deviation tends to increase. This suggests that lower agreement is accompanied by greater variability in annotations, i.e., higher uncertainty among annotators leads to both reduced overlap and less consistent labeling behavior.

2.2 Convert Annotations to Labels

Deriving class labels for every timestep depends on several assumptions about the annotators and the binarisation decision. In our case, we use a weighted majority vote, where the annotations from the creator of the recording are slightly stronger, assuming the creator might have more precise information about the events. The decision boundary for binarisation is set to 50%.

These assumptions give a decent first impression about likely class labels. Given some unclear areas and only 2 annotators, the creator will always be the deciding factor. Given more than 2 annotators, the majority can outvote the creator.

2.3 Label Characteristics

By analysing the class frequency and distribution in the dataset, some interesting patterns emerge. For example, in Figure 4, we can see that although `footsteps` accounts for the highest number of labeled timesteps, the corresponding segments tend to be relatively short. In contrast, `vacuum_cleaner` appears in much longer continuous segments, but only in a small number of occurrences per file. As a result, `vacuum_cleaner` ends up somewhere in the middle of the overall frequency ranking.

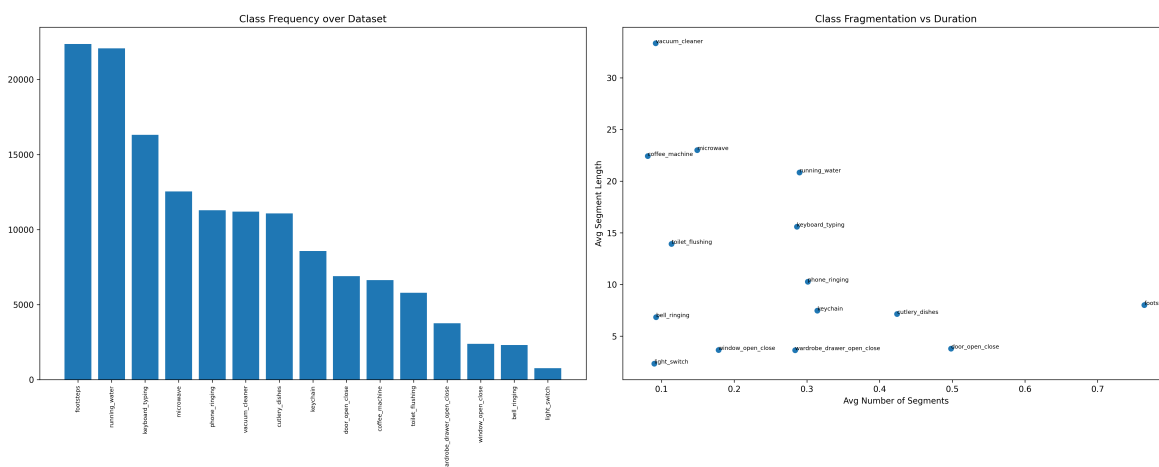


Figure 4: On the left Figure we see the total frequencies of all classes over the entire dataset. On the right the average length of a segment, i.e. an annotated area per label, is plotted against the average amount of such segments in a file.

3 Audio Features

3.1 Metadata

A detailed look into the provided `metadata.csv` reveals several insights regarding the recording properties. We evaluated three specific attributes: `device_placement`, `recording_environment` and `recording_device`. As shown in Figure 5, the device placement is relatively balanced, with 1970 static and 1686 mobile recordings. The most prevalent recording environment is kitchen, followed closely by bedroom and hallway. Notably, 50.8% of the recordings were made using iPhones (any model), which account for 14 of the top 15 recording devices.

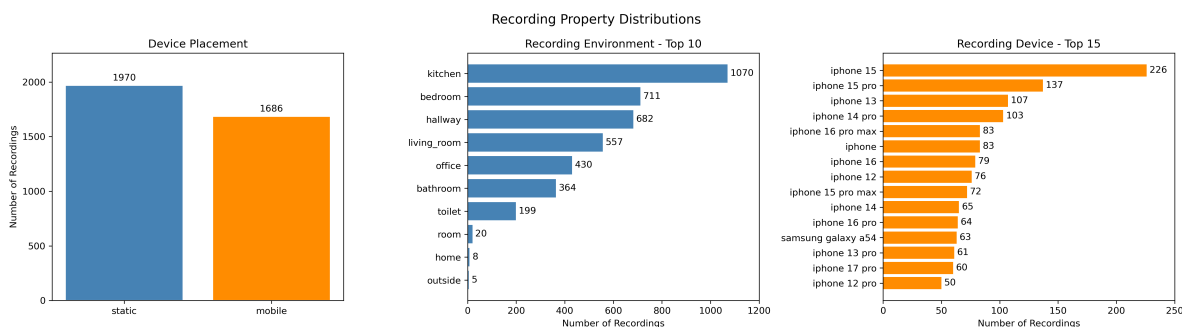


Figure 5: Bar charts for device placement, recording environments and recording devices.

3.2 Feature Statistics

When looking at the scalar feature summary statistics in Table 1, the numbers make sense for everyday household noises recorded on different mobile devices.

For example, the spectral centroid is relatively high (around 2200 Hz) and the flatness is very low (0.088). This shows we captured many bright, structured "ringing" sounds, like those you get from rattling keys or clinking metal cutlery. The most interesting part, however, is the huge gap between the average and maximum volume levels. The mean power is only 63, but the maximum is over 11,000. Big jumps like this are likely caused by two main reasons. First, sudden loud noises — like slamming doors or clattering dishes — cause big spikes in the audio. Second, since everyone used different mobile phones to record, the microphones likely all had very different baseline volumes.

Table 1: Summary statistics of extracted scalar audio features.

Feature	Mean	Std	Min	Max
zcr	0.191	0.133	0.000	0.813
flux	1.953	2.550	0.000	49.975
flatness	0.088	0.087	0.000	1.000
centroid	2207.009	884.199	0.000	5788.952
bandwidth	2784.847	518.917	0.000	4359.974
rolloff_low	299.441	321.524	0.000	3787.188
rolloff_high	5073.701	1327.630	0.000	7514.688
energy	31.684	46.225	0.000	615.892
power	63.393	204.084	0.000	11140.392

3.3 Feature Correlation

In the next step, we examine how different feature categories correlate with one another. We pooled all time segments across all recordings into a single table, where each row is one time segment and each column is one feature category (multi-dimensional features such as MFCCs or Mel-spectrograms are reduced to a scalar by averaging over their feature dimension). We then computed the Pearson correlation matrix between columns (Figure 6) — quantifying how strongly two feature categories co-vary across segments.

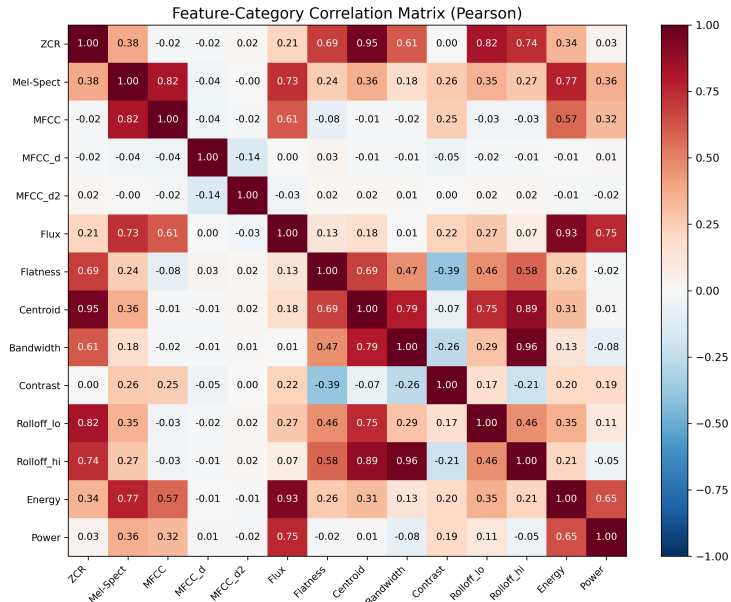


Figure 6: Pearson correlation matrix for all feature categories.

3.4 Feature Space Visualisation

To approach the dimensionality reduction, we decided to only use features that are not highly correlated. After analysing the correlation matrix, we decided to keep the Zero Crossing Rate, MFCC and both its first and second derivation, Contrast and Power. This allowed us to capture the most variance without overloading the model or overemphasising redundant data caused by high correlation. To avoid losing too much information, we retained the individual timesteps instead of collapsing them into samples. This left a matrix of $[T = 100.000 \times \text{Features} = 24]$, which is still far too large to extract any meaningful information from it in a down-projection. Next, we cleaned this data by keeping only those timesteps where exactly one target class was active. Additionally, we required a 100 % agreement among annotators. Still facing $[60.000 \times 24]$ data points, we resorted to kMeans clustering. We retained 5000 clusters and used the most centered point of each cluster for the projections. Trying both PCA and t-SNE - which now worked well after we cut down on samples so much - we decided on using the latter. For t-SNE's perplexity hyperparameter we used the rule of thumb $\frac{\sqrt{N}}{5} = \frac{\sqrt{5.000}}{5} \approx 15$.

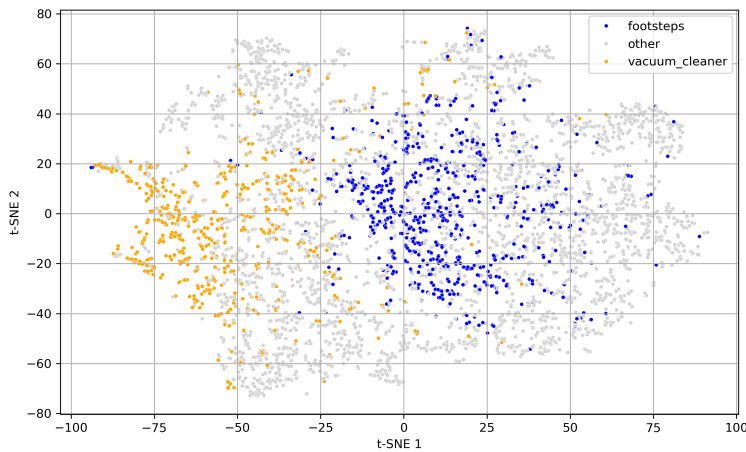


Figure 7: Highlighted are the two classes `vacuum_cleaner` and `footsteps` for their strong distinctions.

In Figure 7 we can see that the classes `vacuum_cleaner` and `footsteps` are well separable. These class clusters are generally compact and show relatively low scattering.

4 Conclusions

4.1 Biases

The dataset's creation process inherently introduced several biases that have to be considered. First, the overwhelming use of personal smartphones (specifically iPhones) introduced a severe hardware bias - meaning the model might learn device-specific audio signatures rather than the actual sounds, which could limit the accuracy when deployed on different microphones. Second, relying exclusively on university students and their household situations leads to a demographic and acoustic bias, which could cause the model to show lower generalisation performance in different settings. Finally, the dataset contains a strong bias towards staged situations which were often recorded unnaturally close - because of that, the model will likely struggle to detect similar events in messy, real-world environments with lower signal-to-noise ratios.

4.2 Conclusion

The very high correlation among most features will pose a challenge for training a model on this data. The projection showed that the target classes are separable, yet not easily so. Annotator differences are present, especially for certain classes like `wardrobe_drawer_open_close`; however, this should be counteracted to a large degree by the annotation review.

4.3 Broader Impact

Publicly releasing this dataset would enable the development of beneficial applications such as assistive technologies for hearing-impaired individuals (e.g., detecting smoke alarms or doorbells) and advance reproducible audio AI research. However, the domestic nature of the recordings poses privacy risks: models trained on this data could be repurposed for secret home surveillance, behavior profiling, or inferring daily routines without residents consent. To mitigate these risks, one could apply a restrictive license prohibiting surveillance use cases, release only pre-computed features rather than raw audio, and publish a datasheet documenting intended and prohibited use cases.

5 Disclosure of LLM and AI Tool Use

For this task ChatGPT-5.3, Gemini-3.1 and Claude Sonnet 4.6 were used to brainstorm as well as to debug code quickly and perform minor spell checking. The correctness was verified by usage of own knowledge. Suggestions are never trusted fully but rather taken as a possibility to be considered.